

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT1 — RELATIONSHIP VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Relationship Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Analyze relationships between variables using scatterplots, correlograms, PCA, bubble charts, and network graphs to construct and interpret appropriate visualizations. (BL4)</i>		
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Section / Batch:		Date:	22.08.2026

Code of Conduct

I Priyanka.G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale plotting.
- Show all supporting calculations (e.g., variance explained, correlation strength, node degree) where relevant.
- Every chart must include a title, correctly labeled axes, and a legend where relevant.

Annexure A – RELATIONSHIP VISUALIZATION LAB

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5 Scatterplot Screen Time and Sleep Hours (5 Marks)

Given Data:

Screen Time (hrs/day)	2	3	4	5	6	7	8	9
Sleep Hours	8.5	8	7.5	7	6.5	6	5.5	5

Task: Construct a scatterplot and identify the type of relationship shown.

Construction Plan (Key Elements):

Chart Type	Scatterplot, with a fitted trend line
X-Axis	Screen Time (hours/day) numeric, quantitative
Y-Axis	Sleep Hours numeric, quantitative
Relationship Type	Strong negative linear relationship

Construction & Analysis:

Checking the slope between every consecutive pair of points: (2,8.5)→(3,8) falls by 0.5; (3,8)→(4,7.5) falls by 0.5; (4,7.5)→(5,7) falls by 0.5; and this exact 0.5-hour decrease in sleep per 1-hour increase in screen time holds for every remaining pair up to (8,5.5)→(9,5). Because the rate of change is identical (−0.5) across the entire dataset, all 8 points fall on a single straight line rather than merely trending in one direction with scatter around it.

This line can be written explicitly as $\text{Sleep Hours} = 9.5 - 0.5 \times (\text{Screen Time})$, which can be verified against the data: at Screen Time = 2, predicted Sleep = $9.5 - 0.5(2) = 8.5$ ✓; at Screen Time = 9, predicted Sleep = $9.5 - 0.5(9) = 5.0$ ✓. Every one of the 8 data points satisfies this equation exactly.

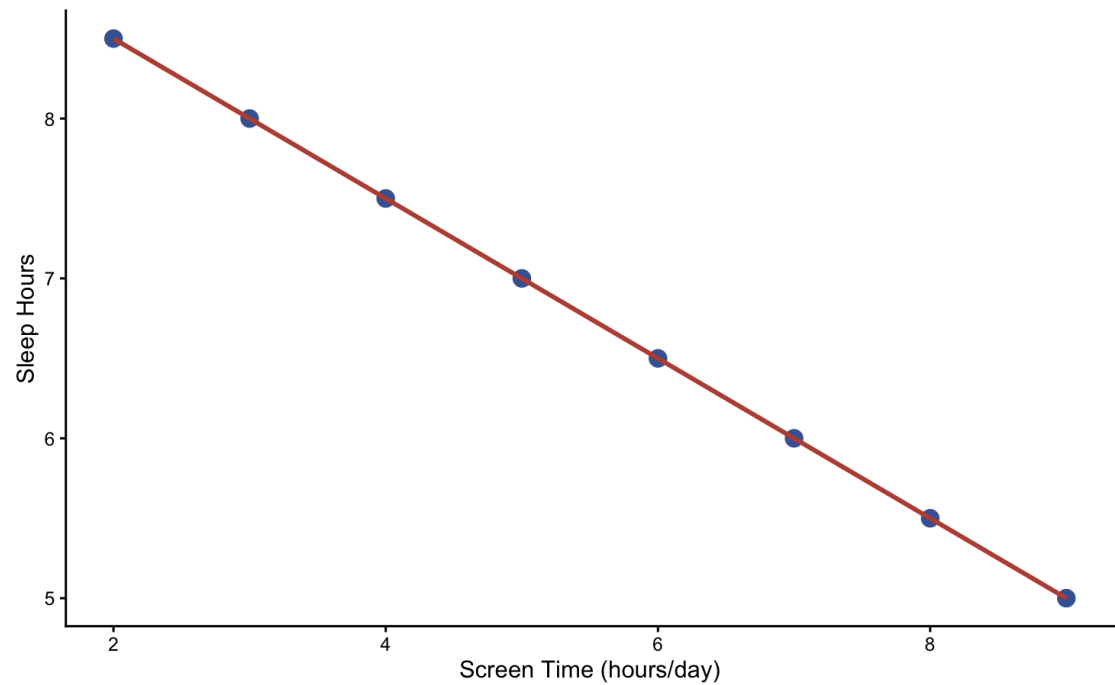
Because the relationship is both linear (constant slope) and negative (sleep decreases as screen time increases) with essentially perfect fit (all points on the line, correlation $r \approx -1$), it is classified as a strong negative linear relationship as screen time rises, sleep hours fall at a constant, predictable rate, with no evidence of curvature or diminishing/accelerating effects anywhere in the observed range (2 to 9 hours of screen time).

A scatterplot is the correct chart for establishing this relationship in the first place, since it is the standard technique for examining how two continuous numeric variables (Screen Time and Sleep Hours) relate to one another across individual observations; a bar chart or line chart connecting the teenagers in an arbitrary order would not reveal the underlying functional relationship the way plotting one variable directly against the other does.

Constructed Chart:

192524124 Problem 5 - Screen Time and Sleep Hours

Scatterplot showing the relationship between daily screen time and sleep



Interpretation: The scatterplot shows a strong negative linear relationship. As screen time increases, sleep hours decrease.

Figure 5: Scatterplot Screen Time vs. Sleep Hours, with Fitted Trend Line

Problem 6 Correlogram Height, Weight, and Age (5 Marks)

Given Data:

Pair	Correlation (r)
Height Weight	0.85
Height Age	0.15
Weight Age	0.30

Task: Construct a correlogram (describe the shading/color for each cell) and identify the strongest pair.

Construction Plan (Key Elements):

Chart Type	Correlogram (3×3 correlation heatmap)
Colour Scale	Sequential, red (weak, ≈ 0) → orange/yellow (moderate) → dark blue (strong, ≈ 1)
Diagonal Cells	Always 1.00 (dark blue) every variable is perfectly correlated with itself
Strongest Pair	Height Weight, $r = 0.85$

Construction & Analysis:

The correlogram is a 3×3 grid where both axes list the same three variables (Height, Weight, Age), and each cell's fill colour encodes the correlation between the row and column variable using a sequential colour scale: the three diagonal cells (Height–Height, Weight–Weight, Age–Age) are always exactly 1.00 by definition and are shaded the darkest blue, since a variable is always perfectly correlated with itself.

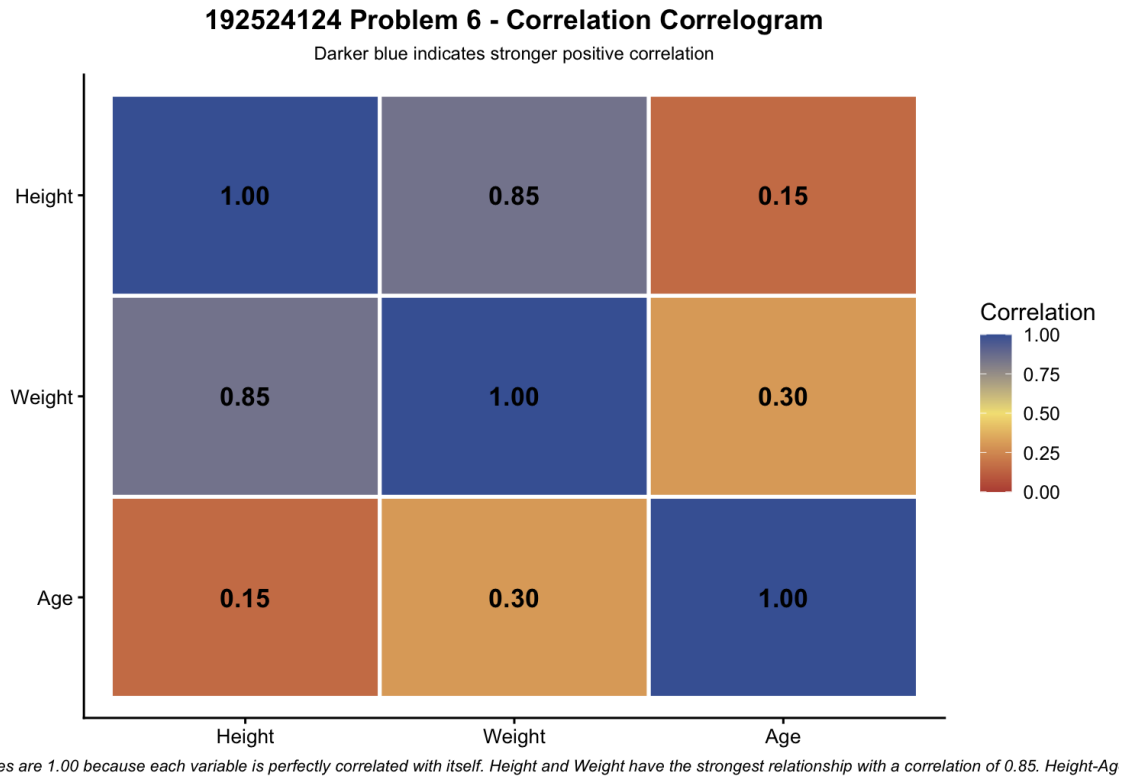
The off-diagonal cells are shaded according to their correlation value: Height–Weight = 0.85 is shaded a medium-dark blue-grey, reflecting a strong positive correlation close to the diagonal's maximum; Weight–Age = 0.30 is shaded a light orange/yellow, reflecting a weak-to-moderate positive correlation; and Height–Age = 0.15 is shaded the most reddish-orange of the three off-diagonal cells, reflecting the weakest correlation in the dataset, closest to zero.

The strongest pair is Height–Weight, with $r = 0.85$, which is more than five times larger than Height–Age (0.15) and nearly three times larger than Weight–Age (0.30). This makes physiological sense: taller individuals in a sample tend to also weigh more, so Height and Weight move together closely. By contrast, Height–Age (0.15) and Weight–Age (0.30) are both much weaker, suggesting that within this sample, a person's Age is only loosely associated with their Height or Weight consistent with the fact that, unlike Height and Weight, Age does not have a direct physiological mechanism linking it to body size once past a certain developmental stage.

A correlogram is the appropriate technique here specifically because there are three separate pairwise relationships to compare at once (Height–Weight, Height–Age, Weight–Age); a single scatterplot can only show one pair of variables at a time, so a three-variable dataset would require three separate scatterplots to convey the same information a single correlogram grid conveys in one glance, with the added benefit that colour intensity lets all

three comparisons be ranked by strength immediately, without needing to visually estimate three different point-cloud shapes.

Constructed Chart:



as are 1.00 because each variable is perfectly correlated with itself. Height and Weight have the strongest relationship with a correlation of 0.85. Height-Ag

Figure 6: Correlogram Height, Weight, and Age

Problem 12 Scatterplot & PCA Math and Science Scores (5 Marks)

Given Data:

Student	1	2	3	4	5
Math Score	60	70	80	90	50
Science Score	65	72	78	85	55

Task: Plot the scatter and explain whether PC1 would align closely with the diagonal of the plot, and why.

Construction Plan (Key Elements):

Chart Type	Scatterplot with a reference diagonal ($y = x$ style line)
Technique	Principal Component Analysis (PCA) direction of maximum variance
PC1 Interpretation	The axis along which the data varies the most
Conclusion	PC1 aligns closely with the diagonal, since Math and Science move together

Construction & Analysis:

Mean Math score = $(60+70+80+90+50) \div 5 = 350 \div 5 = 70$. Mean Science score = $(65+72+78+85+55) \div 5 = 355 \div 5 = 71$.

Checking each student's deviation from these means: Student 1 is $(60-70, 65-71) = (-10, -6)$; Student 2 is $(70-70, 72-71) = (0, +1)$; Student 3 is $(80-70, 78-71) = (+10, +7)$; Student 4 is $(90-70, 85-71) = (+20, +14)$; Student 5 is $(50-70, 55-71) = (-20, -16)$. In every single case, the Math deviation and Science deviation share the same sign (both negative, or both positive) and are of broadly similar magnitude students who score below average in Math also score below average in Science, by a comparable margin, and likewise above average in both together.

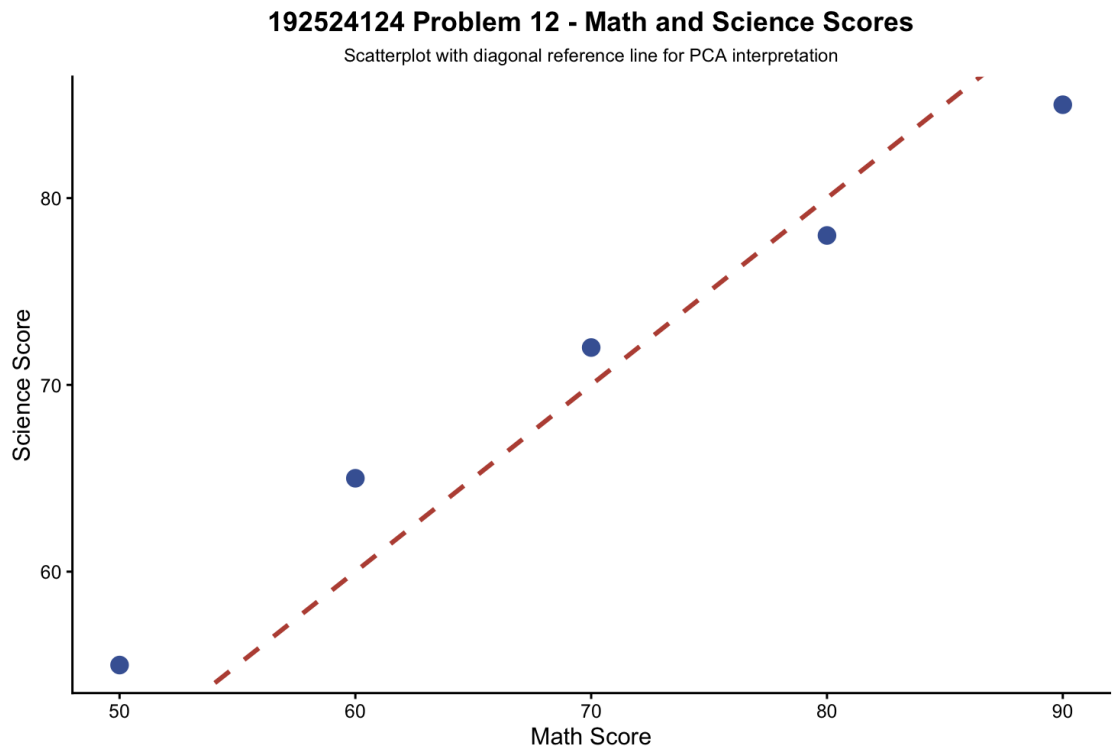
PC1, by definition, is the direction in the data that captures the greatest amount of variance. Because every student's Math and Science deviations move together in the same direction, almost all of the spread in this dataset lies along the 'both scores rise or fall together' direction which is exactly the diagonal (the line where Math Score = Science Score, approximately). PC1 would therefore align closely with this diagonal, since that is the direction along which the five points are most spread out.

By contrast, PC2 (the direction orthogonal to PC1) would capture the much smaller remaining variation specifically, each student's individual tendency to be relatively stronger in one subject than the other (e.g., Student 1 is relatively weaker in Math than Science given their deviations of -10 vs. -6). Since this 'difference between subjects' variation is consistently small compared to the 'overall performance level' variation, PC1 dominates and sits close to the diagonal, exactly as shown by how tightly the five points hug the dashed reference line in the constructed plot.

This generalises to a broader PCA principle: whenever two variables are strongly and positively correlated and measured on comparable scales, the scatter of points forms an elongated cloud stretched along the shared-direction diagonal rather than a circular cloud, and PCA's first component always aligns with the direction of greatest elongation so strong

positive correlation between two variables is itself a reliable predictor that PC1 will sit close to the diagonal, without needing to compute the full eigen-decomposition to anticipate the result.

Constructed Chart:



Interpretation: The points follow a strong positive diagonal pattern. PC1 would align closely with this diagonal because Math and Science scores increase together.

Figure 12: Scatterplot of Math vs. Science Scores with Diagonal Reference Line

Problem 18 Bubble Chart Product Price, Rating, and Sales (5 Marks)

Given Data:

Product	Price (\$)	Rating (/5)	Units Sold
P1	20	4.5	500
P2	35	3.8	300
P3	15	4.2	700
P4	50	4.8	150
P5	25	4	600
P6	40	3.5	250

Task: Construct a bubble chart and identify which product has the best combination of high rating and high sales.

Construction Plan (Key Elements):

Chart Type	Bubble chart (3-variable encoding)
X-Axis	Price (\$) numeric, quantitative
Y-Axis	Customer Rating (/5) numeric, quantitative
Bubble Size	Units Sold numeric, quantitative (third variable)

Construction & Analysis:

Ranking all six products by Customer Rating (highest to lowest): P4 (4.8) > P1 (4.5) > P3 (4.2) > P5 (4.0) > P2 (3.8) > P6 (3.5).

Ranking the same six products by Units Sold (highest to lowest): P3 (700) > P5 (600) > P1 (500) > P2 (300) > P6 (250) > P4 (150).

P4 has the single highest rating (4.8) but by far the lowest sales (150 units) the largest rating-vs-sales mismatch in the dataset, making it a low-volume, premium-niche product rather than an overall strong performer. P3, by contrast, ranks 3rd in rating (4.2, still well above the midpoint of the 3.5–4.8 range) while ranking 1st in sales (700 units, nearly five times P4's volume) the strongest joint performance across both dimensions of any product in the set.

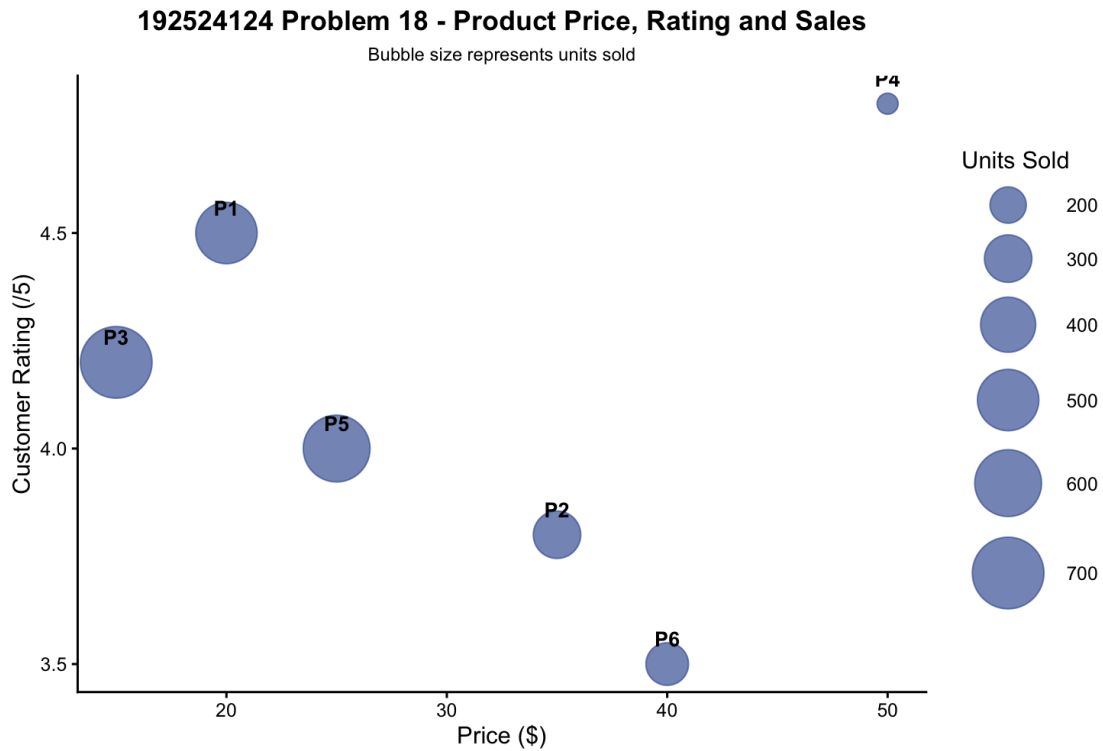
- P3: Price \$15 (lowest), Rating 4.2 (3rd highest), Units Sold 700 (highest) best combined performer.
- P1: Price \$20, Rating 4.5 (2nd highest), Units Sold 500 (3rd highest) a strong runner-up.
- P4: Price \$50 (highest), Rating 4.8 (highest), Units Sold 150 (lowest) high quality, but low-volume.

P3 is therefore identified as having the best combination of high rating and high sales: visually, it is both positioned high on the Y-axis (strong rating) and drawn as the largest bubble in the chart (highest sales), a combination no other product matches P1 comes closest on rating and P5 comes closest on sales, but neither leads on both dimensions simultaneously the way P3 does.

A bubble chart is the correct technique for this three-variable comparison because it encodes Price and Rating using position (the two axes, which the eye reads most precisely)

while encoding the third variable, Units Sold, using bubble area allowing all three variables to be read from a single mark per product, without needing a separate chart or a colour-based third encoding that would be harder to compare quantitatively than size once several bubbles are on screen at once.

Constructed Chart:



Interpretation: P3 has the highest sales and a high rating, giving it the best overall combination of customer rating and sales volume.

Figure 18: Bubble Chart Price vs. Rating, Bubble Size = Units Sold

Problem 24 Network Graph Researcher Collaboration (5 Marks)

Given Data:

Edge 1	Edge 2	Edge 3	Edge 4	Edge 5
R1–R2	R1–R3	R2–R3	R4–R5	R5–R6

Task: Draw the network graph and determine whether it forms one connected group or multiple separate groups.

Construction Plan (Key Elements):

Chart Type	Network (node-link) graph
Nodes	6 researchers: R1, R2, R3, R4, R5, R6
Edges	5 collaboration links, as listed in the given data
Key Reading	A connected component = a set of nodes all reachable from one another via edges

Construction & Analysis:

Building the adjacency from the 5 given edges: R1–R2, R1–R3, and R2–R3 connect R1, R2, and R3 to one another in every possible pairing this is a complete triangle (a 3-node clique), meaning each of these three researchers has directly collaborated with both of the other two.

Separately, R4–R5 and R5–R6 connect R4, R5, and R6 into a chain, with R5 sitting at the centre as the only researcher directly connected to both R4 and R6 R4 and R6 are not directly connected to each other, but are still reachable via R5.

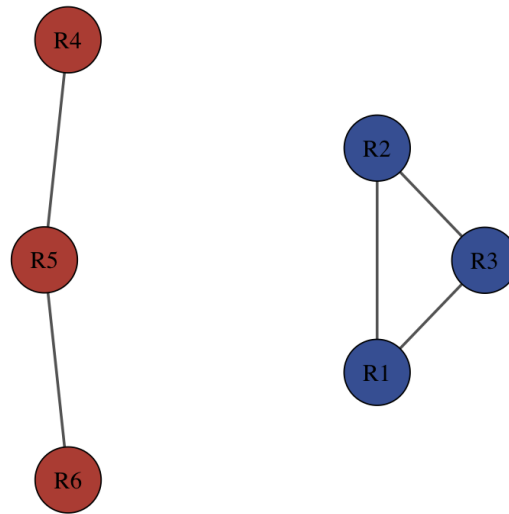
Checking for any edge that crosses between {R1, R2, R3} and {R4, R5, R6}: none of the 5 given edges connects a researcher in the first group to a researcher in the second group. Since there is no path direct or indirect from any node in {R1, R2, R3} to any node in {R4, R5, R6}, the graph is disconnected and forms exactly two separate connected groups: Group 1 = {R1, R2, R3} and Group 2 = {R4, R5, R6}.

The two groups also differ structurally, not just in membership: Group 1 is a complete triangle, meaning collaboration is maximally dense every researcher has worked directly with every other researcher in that group. Group 2 is a simple chain, meaning collaboration is less dense R5 is a critical bridge, and if R5's collaboration link were removed, Group 2 would fracture further into two fully isolated researchers (R4 and R6), whereas Group 1 has no single point of failure of that kind, since removing any one edge still leaves the remaining two researchers connected via the third.

A network (node-link) graph is the correct technique for this data because the underlying structure is fundamentally relational the question is not about comparing six individual quantities, but about which pairs of researchers are directly linked and whether those links chain together into larger groups; a bar or scatter chart has no natural way to encode 'connection between two specific entities' the way an edge drawn between two nodes does, which is exactly the information this task requires (determining connected components).

Constructed Chart:

192524124 Problem 24 - Researcher Collaboration Network



Interpretation: The network contains two separate connected groups.

Group 1 consists of R1, R2 and R3.

Group 2 consists of R4, R5 and R6.

Therefore, the network is disconnected and forms multiple separate groups.

Figure 24: Network Graph Two Disconnected Collaboration Groups